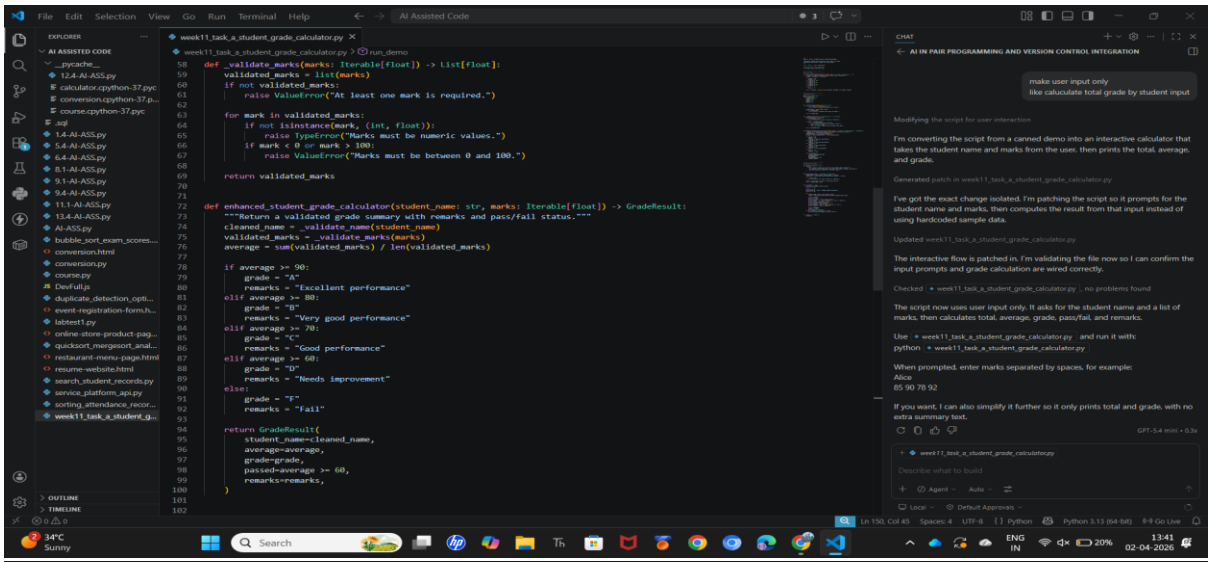
21.5-AI-ASS
2303A52249
A.VAMSHI KRISHNA

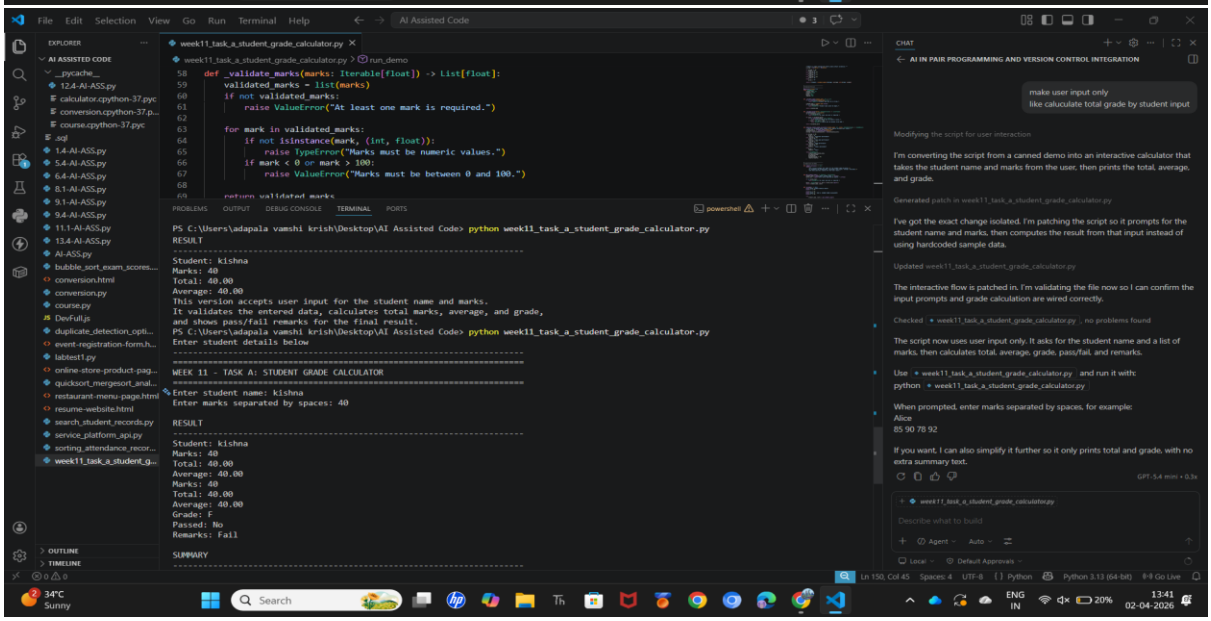
TASK-1

Prompt: AI-Based Initial Program Development

Use AI to generate a basic program (e.g., Student Grade Calculator).
Ask AI to enhance it with validation and additional features. Compare both versions.

Code & Output:





Observation:

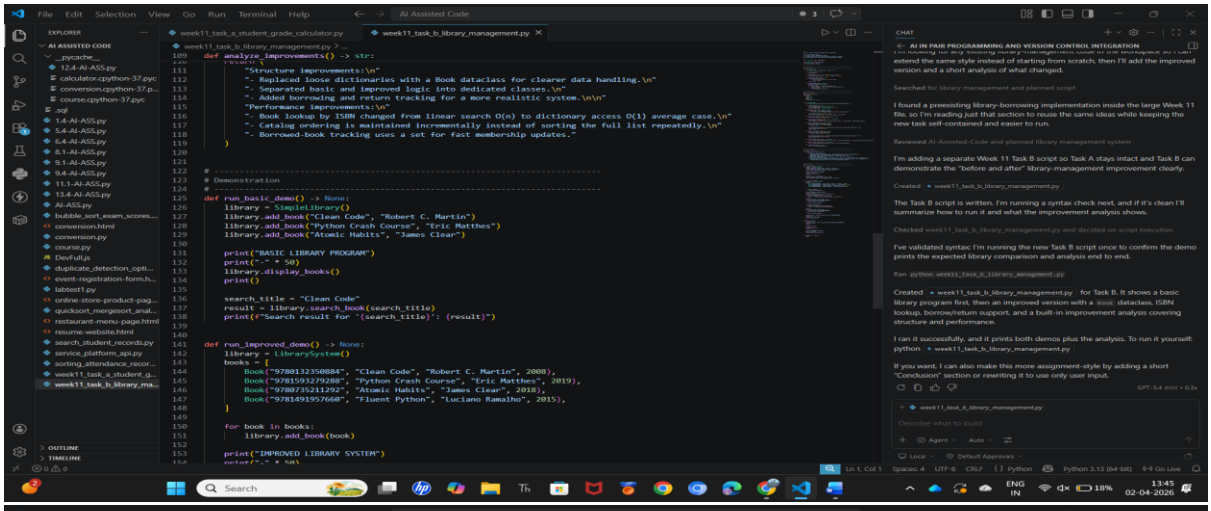
The AI-generated initial program provided a quick and correct baseline implementation. After enhancement, the program included input validation and additional features, improving reliability and usability. The comparison showed that AI significantly helps in refining logic and reducing development time.

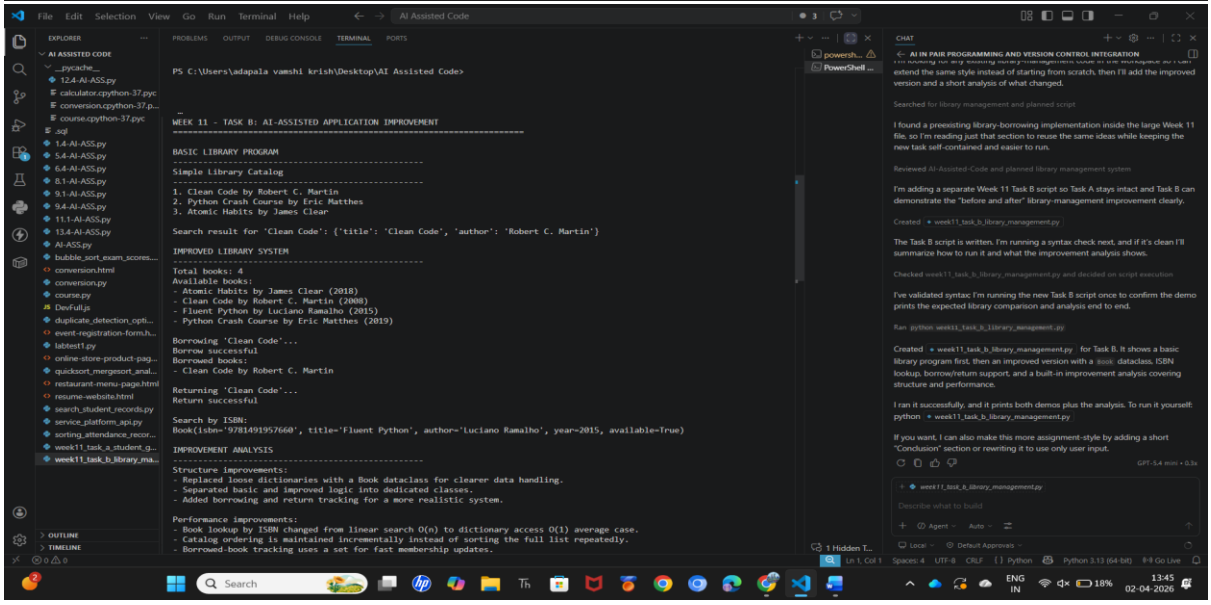
TASK-2

Prompt: AI-Assisted Application Improvement

Generate a simple program (e.g., Library Management).
Use AI to add new functionalities.
Analyze improvements in structure and performance.

Code & Output:





```
PS C:\Users\adapala vamshi krish\Desktop\AI Assisted Code>

WEEK 11 - TASK B: AI-ASSISTED APPLICATION IMPROVEMENT
============================================================

BASIC LIBRARY PROGRAM
------------------------------------------------------------
Simple Library Catalog
------------------------------------------------------------
1. Clean Code by Robert C. Martin
2. Python Crash Course by Eric Matthes
3. Atomic Habits by James Clear

Search result for 'Clean Code': {'title': 'Clean Code', 'author': 'Robert C. Martin'}

IMPROVED LIBRARY SYSTEM
------------------------------------------------------------
Total books: 4
Available books:
- Atomic Habits by James Clear (2018)
- Clean Code by Robert C. Martin (2008)
- Fluent Python by Luciano Ramalho (2015)
- Python Crash Course by Eric Matthes (2019)

Borrowing 'Clean Code'...
Borrow successful
Borrowed books:
- Clean Code by Robert C. Martin

Returning 'Clean Code'...
Return successful

Search by ISBN:
Book(isbn='9781491957660', title='Fluent Python', author='Luciano Ramalho', year=2015, available=True)

IMPROVEMENT ANALYSIS
------------------------------------------------------------
Structure improvements:
- Replaced loose dictionaries with a Book dataclass for clearer data handling.
- Separated basic and improved logic into dedicated classes.
- Added borrowing and return tracking for a more realistic system.

Performance improvements:
- Book lookup by ISBN changed from linear search O(n) to dictionary access O(1) average case.
- Catalog ordering is maintained incrementally instead of sorting the full list repeatedly.
- Borrowed-book tracking uses a set for fast membership updates.
```

Observation:

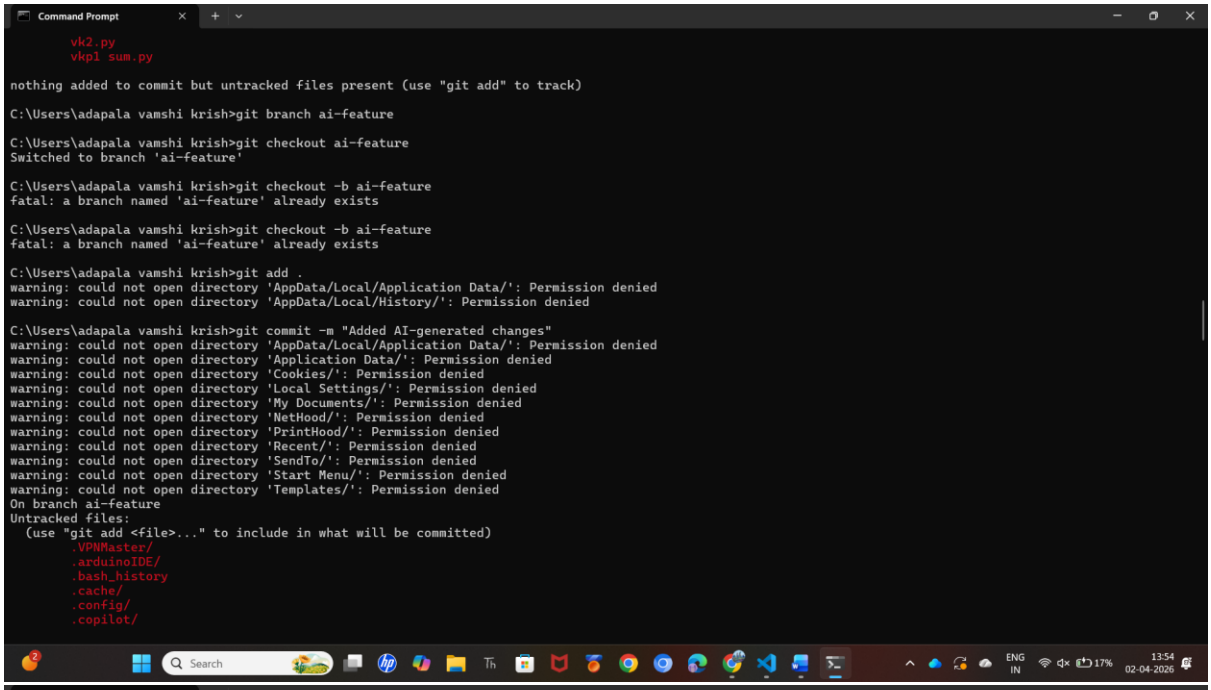
AI-assisted modifications enhanced the structure and functionality of the application. New features were added efficiently, and the code became more modular and readable. Performance and maintainability improved due to better organization and optimized logic.

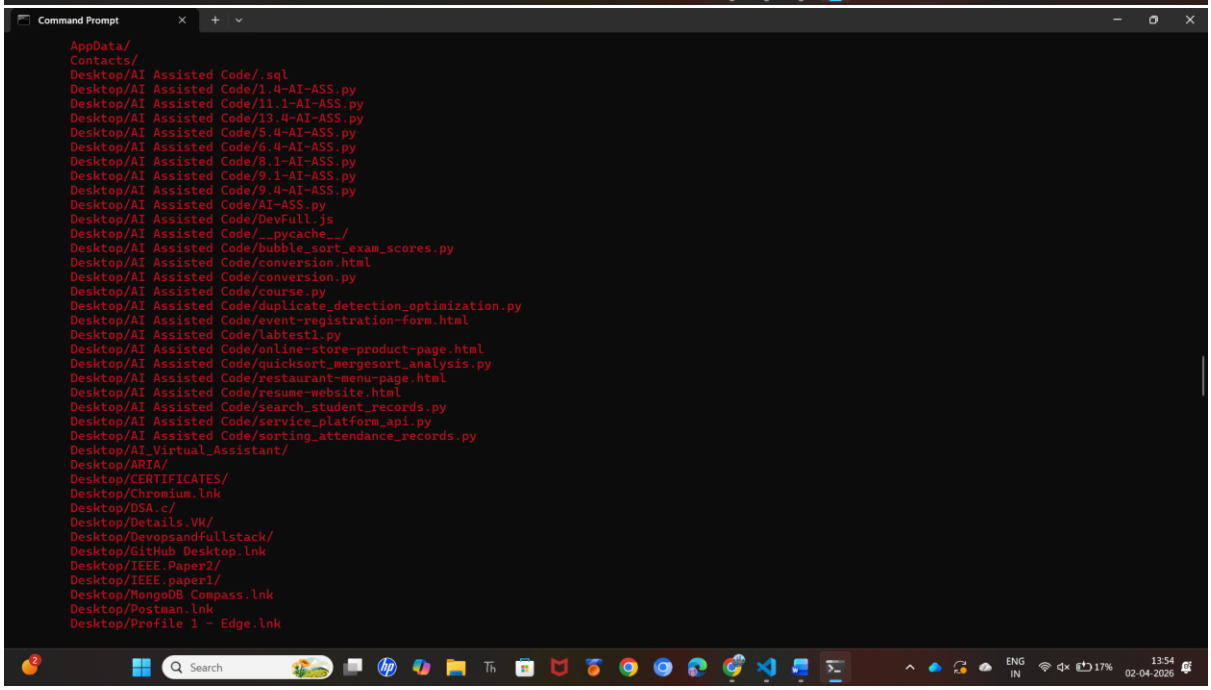
TASK-3

Prompt: Git Workflow Practice

Initialize a Git repository.
Create a new branch and add AI-generated changes.
Commit and merge changes into the main branch.

Code & Output:





Observation:

Git Workflow Practice (Observation)

The Git workflow was successfully implemented by initializing a repository, creating a branch, and merging changes. AI-generated updates were integrated smoothly, demonstrating how version control supports safe and organized development practices.

TASK-4

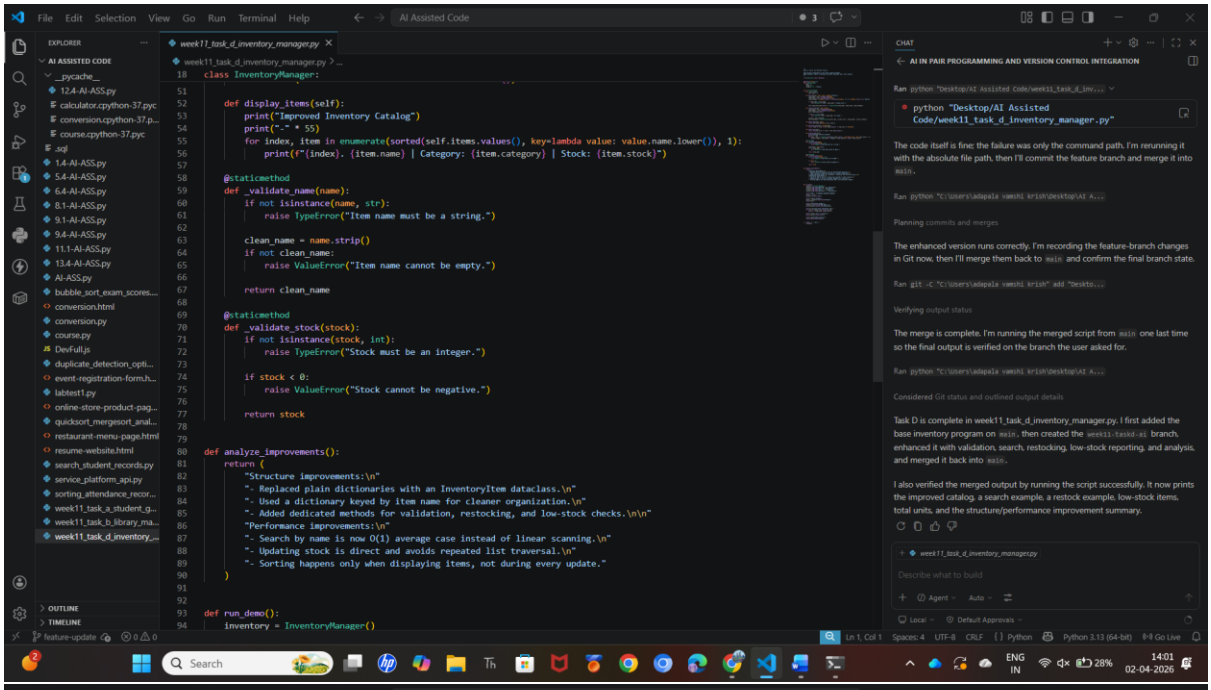
Prompt: Branching and Merging with AI Support

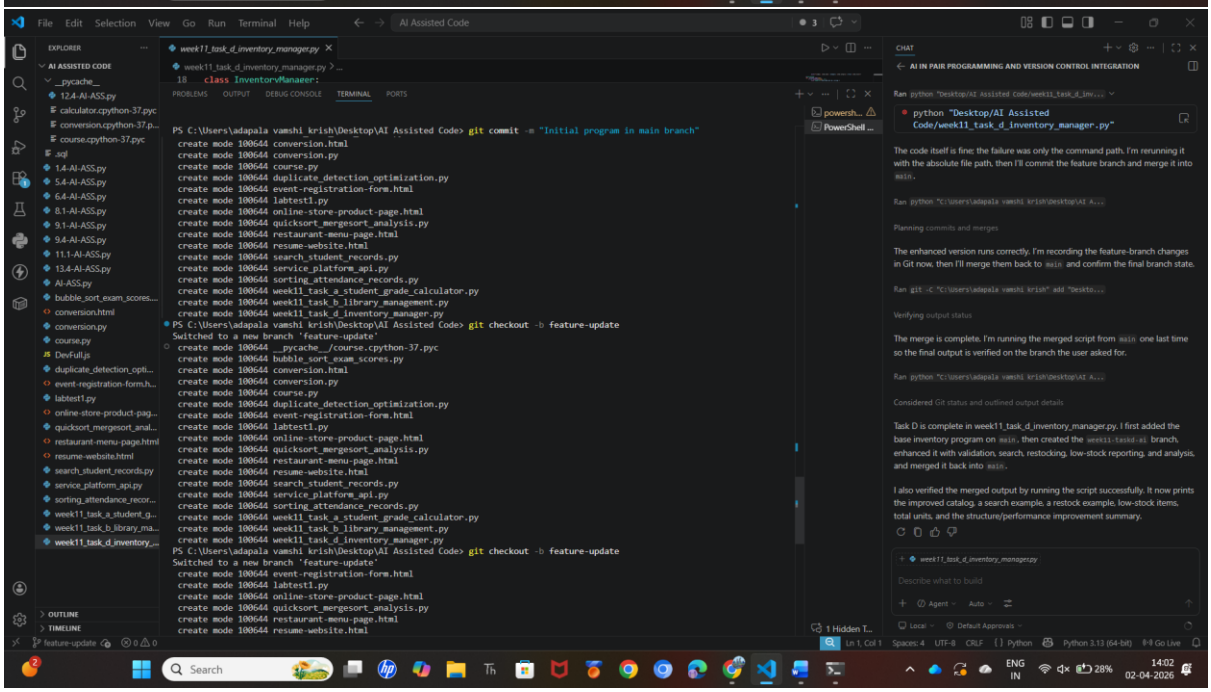
Develop a program in the main branch.

Create a feature branch and modify using AI.

Merge and verify output.

Code & Output:





Observation:

Branching and Merging with AI Support (Observation)

The program developed in the main branch was enhanced in a feature branch using AI. The merge process was completed without conflicts, and the final output verified correctness. This task highlighted effective collaboration between AI assistance and version control workflows.

TASK-5

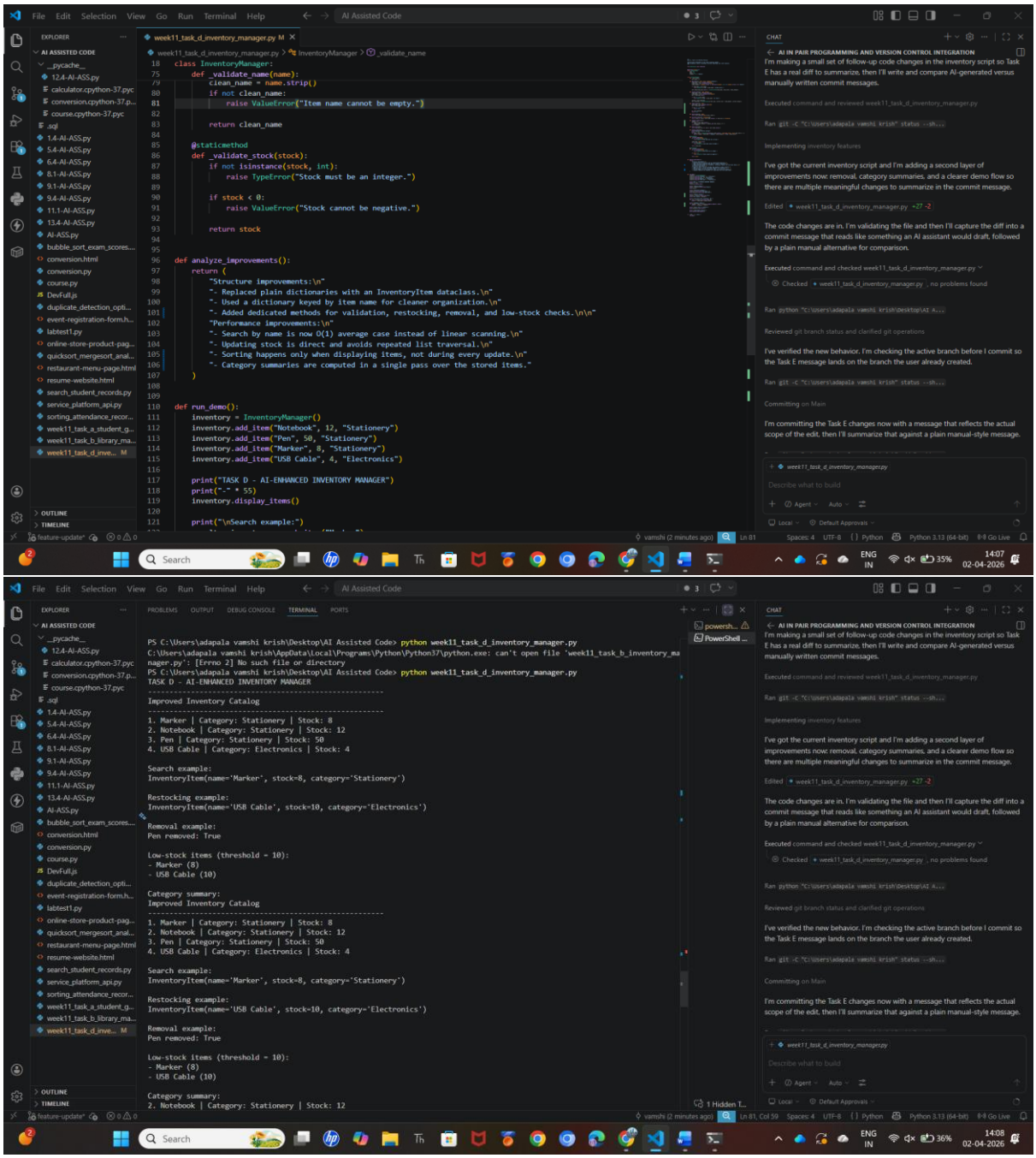
Prompt: AI for Commit Message Writing

Perform multiple code changes.

Use AI to generate commit messages.

Compare with manually written messages.

Code & Output:



Observation:

AI for Commit Message Writing (Observation)

AI-generated commit messages were clear, structured, and descriptive compared to manual ones. They improved readability and documentation of changes. This demonstrated that AI can standardize commit practices and enhance team collaboration.